PRECALCULUS · UNIT 6: MATRICES

Spy Agency Codes

Use matrices to manage inventory, encode secret messages, and solve systems.

CP1

2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Welcome to the Cipher Division. Agents carry gadgets from three warehouses, field reports must be encoded so enemies cannot read them, and supply problems arrive as systems of equations.

You will use matrices to track gadgets, encode and decode messages, and solve supply problems.



Key Matrix

K = [[3, 1], [5, 2]]

Alphabet

A = 1, B = 2, …, Z = 26

space = 27

Learning Goals

- I can perform matrix operations in context.
- I can find determinants and inverses of 2 × 2 matrices.
- I can encode, decode, and solve systems with matrices.

Project Timeline

When	What you will do
Day 1	Inventory & Value
Day 2	Encode & Decode
Day 3	Supply Systems
Final	Agent Mission File + reflection

Part 1: Inventory & Value

Day 1 · Use the Gadget Inventory.

Gadget Inventory (rows: warehouses, columns: gadgets)

Boston	London	Tokyo
Pens 12 · Watches 5	Pens 8 · Watches 9	Pens 15 · Watches 6
Drones 3	Drones 4	Drones 7

Costs	New Shipment	Key Inverse
Pen $40 · Watch $250	Each warehouse gets	$K^{-1} = [[2, -1], [-5, 3]]$
Drone $900	+2 pens, +1 watch, +1 drone	

1. Find I + S and the value of each warehouse after the shipment.

2. Prices rise 10%. Write the new cost matrix and find the new Tokyo value for the original inventory.

3. Is I · C the same as C · I? Explain using dimensions.

4. Find the total of each gadget across all warehouses using [1, 1, 1] · I.

Part 2: Encode & Decode

Day 2 · Key K = [[3, 1], [5, 2]], A = 1, …, Z = 26, space = 27.

5. Find det K and K^{-1}.

6. Encode the message GO NOW in pairs (GO, _N, OW).

7. Decode the message [58, 97], [23, 40].

8. Why must the key have a nonzero determinant?

Part 3: Supply Systems

Day 3 · Use inverse matrices to solve.

9. A shipment of pens and watches has 30 items and costs $3,510. Write a matrix equation and solve it with an inverse matrix.

10. Check your answer by substitution.

11. Write your mission file with inventory, a coded message for a partner, and the system you solved.

Final Product: Agent Mission File

Your final product must include:

- Inventory operations and values
- Determinant and inverse
- Encoded message and decoding
- Matrix system solved and checked
- Mission file

Reflection

R1. What does the determinant tell you about a matrix?

R2. How is solving AX = B with an inverse like solving ax = b?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Matrix operations, inverses, and determinants are correct and used for inventory, codes, and systems. Goes beyond the minimum with a creative, realistic choice.	Matrix operations, inverses, and determinants are correct and used for inventory, codes, and systems.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Mission file includes matrices, coded messages, solved systems, and explanations. Polished, organized, and visually engaging.	Mission file includes matrices, coded messages, solved systems, and explanations. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.